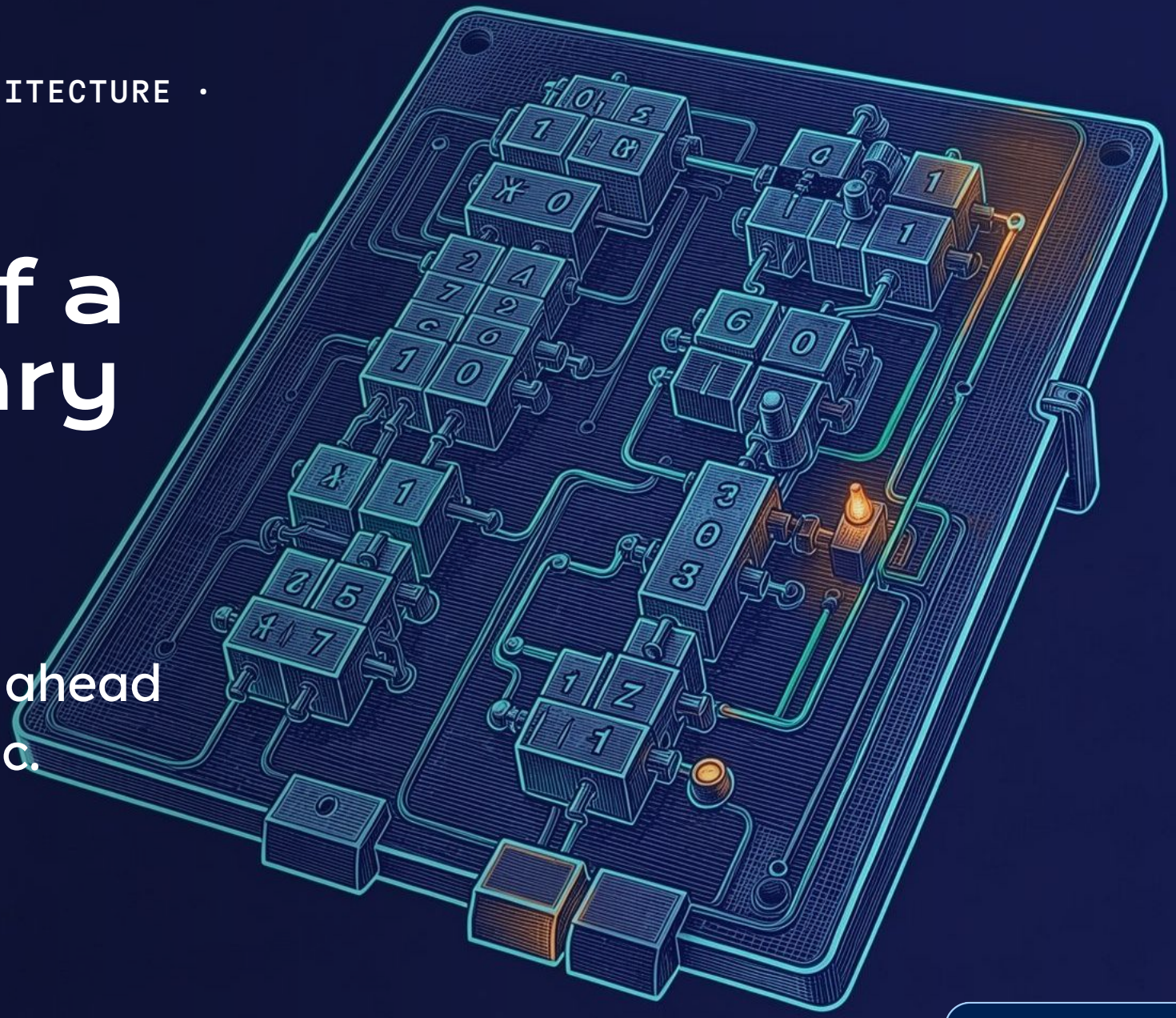
DIGITAL DESIGN AND COMPUTER ARCHITECTURE · MODULE 6

Design and Simulation of a Four-Bit Binary Arithmetic Calculator

Build it, extend it, and look ahead through combinational logic.



Made with GAMMA

WHY THIS CIRCUIT MATTERS

Binary arithmetic is the foundation of digital processors

A four-bit calculator accepts two binary inputs and produces an arithmetic result using combinational logic.

Small arithmetic circuits are building blocks of ALUs and digital processors.



Design

Use logic gates and full-adder blocks.



Implement

Implement arithmetic with logic gates; connect inputs, outputs, and carry.



Verify

Simulate in Logisim Evolution; verify outputs.



Extend

Extend small circuits into larger computer systems.

MODULE 6 CONNECTION

Small circuits can be extended into larger computer systems.

Made with GAMMA

CORE BUILDING BLOCK

A ripple-carry adder builds a four-bit result from full adders

Four bits represent unsigned values from 0000 to 1111. Combinational outputs depend on the current inputs.

FULL-ADDER EQUATIONS

Sum = A XOR B XOR Cin

Carry = (A AND B) OR (Cin AND (A XOR B))

Four cascaded full-adder stages

Carry ripples from the least-significant stage to the final Cout.

Stage 0	Stage 1	Stage 2	Stage 3
A0 · B0	A1 · B1	A2 · B2	A3 · B3
Cin → C1	C1 → C2	C2 → C3	C3 → Cout
Sum S0	Sum S1	Sum S2	Sum S3

Logic vocabulary: AND · OR · XOR · NOT

Future extension: add operation controls and additional arithmetic functions.

Made with GAMMA

SIMULATION WORKFLOW

Logisim Evolution turns the logic into a testable circuit

Input switches feed the logic. Full-adder blocks produce output bits, LEDs display the sum, and a carry indicator shows when the result needs an additional bit.



01

Input switches

Select A and B.

02

Adder stages

Propagate carry.

03

LED outputs

Read S0–S3 and Cout.

CURRENT DESIGN

Four-bit combinational addition.

Made with GAMMA

EVIDENCE OF CORRECTNESS

Truth-table checks make each arithmetic result verifiable

Change input switches, observe LEDs, and compare circuit outputs with expected results to verify and debug.

Evidence boundary. Actual Logisim screenshots are placeholders unless supplied. Do not claim testing without evidence.

TEST CASES

01 · 0011 + 0010 = 0101 · 3 + 2 = 5

02 · 0111 + 0001 = 1000 · 7 + 1 = 8

03 · 1111 + 0001 = 1 0000 · five bits, so Cout = 1.

FULL-ADDER TRUTH TABLE

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Made with GAMMA

LOOK AHEAD

The calculator is a small step toward extensible arithmetic hardware

CORE AND CURRENT DESIGN

Four-bit combinational addition uses input switches, ripple carry, sum LEDs, and a carry indicator. Combinational logic provides the arithmetic core used in ALUs, processors, embedded systems, and digital calculators.

FUTURE EXTENSIONS

Subtraction, operation selection, seven-segment display, operand/result registers, and a simplified RISC-V datapath.

PERFORMANCE CONCEPTS

Clock period, CPI, pipelining, caching, and parallelism.

MODULE 6 BRIDGE

CPU instructions, control signals, and memory-mapped outputs can select ALU operations and drive LEDs or displays.

BROADER PATHWAYS

Embedded systems, IoT, CMOS VLSI, TinyML, and edge-AI hardware.

SCOPE

Future extensions and applications, not implemented features of the current calculator.

Made with GAMMA

DIGITAL DESIGN AND COMPUTER ARCHITECTURE

THANK YOU

Design and Simulation of a Four-Bit Binary Arithmetic Calculator Using Combinational Logic



Made with GAMMA